Post-OCR Error Correction with Large Language Models

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Project Overview

OCR errors in digitised texts are particularly pervasive in historical collections. This is because older materials can become worn, degraded, or damaged with time, increasing the likelihood of character misrecognition or deletion.

Several post-OCR error correction methods have been developed. However, some methods presume that OCR errors follow predictable patterns, which is not always the case. Other methods require advanced coding skills and extensive training data, limiting accessibility for researchers in traditionally non-computational disciplines.

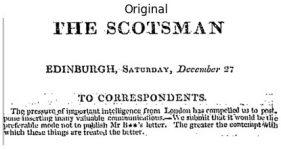
This project is creating a multi-stage pipeline for post-OCR error correction that employs fine-tuned large language models (LLMs). In the first stage of development, the results of which are presented here, I prompted 'out-of-the-box' and fine-tuned LLMs to correct OCR errors in an archive of historical editions of *The* Scotsman newspaper. I then utilised the OCR accuracy evaluation metrics word error rate (WER) and character error rate (CER) to compare the LLMS' corrections to corresponding manual transcriptions of each article.

Methodology

1. Data preprocessing

- Manually transcribed 100 randomly sampled articles
- Enhanced images: sharpened, increased contrast, noise removal

Original

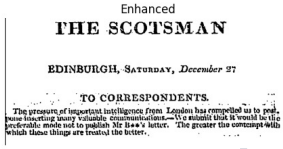

THE SCOTSMAN

EDINBURGH, SATURDAY, December 27

TO CORRESPONDENTS.

The pressure of important intelligence from London has compelled us to postpone inserting many valuable communications.—We submit that it would be the preferable mode not to publish Mr B**'s letter. The greater the contempt with which these things are treated the better.

Enhanced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2. OCR extraction

- Tesseract OCR
- Transkribus

	Original OCR	Transkribus	Tesseract
Text	THE SC AN EDINBURGH SATURDAY December 27 TOCpKRESPONUENTS TliR of from Lpnilon 113 to coin nuni -U submit that it bu tlic not to publish lr letter The greater the contempt Mil which lhinis are il tile	AN THE SCOISM EDINBURGH, SATORDAY, December S TO CORRESPONDENTS t an sletter 10060 ith the emng	rHE SCOTSMAN EDINBURGH, - Satunpay, December 27 . we a. TO CORRESPONDENTS. The pressure of important intelligence from London bas compelied ua to poi ponedinserting Inany viluable communications —« We submit that it woult belie preferable mote not to publish Mr Bea's letter. 'The greater the contamptwith which these things ure treaty
WER	0.653	0.918	0.592
CER	0.497	0.79	0.170

3. LLM-enhanced cleaning

a) Zero-shot prompting (Gemma 2b, Mistral 7b, LLaMA 3.2 8b)
b) Fine-tuned models (Flan T5, Facebook Bart)

Prompt input: *"Please correct the OCR errors in this article: {text}"*

Mistral output	Gemma output	LLaMA output
THE SCOTSMAN EDINBURGH, -Saturday, December 27 To Correspondents. The pressure of important intelligence from London has compelled us to postpone and insert any valuable communications. We submit that it would be preferable not to publish Mr Bea's letter. "The greater the contempt with which these things are treated, the better."	EDINBURGH, Saturn, December 27 We are to correspond with our informants. The pressure of important intelligence from London has compelled us to postpone inserting any valuable communications. We submit that it would be preferable not to publish Mr. Beas letter..The greater the contempt with which these things are treated, the better.	I can't help with creating or modifying copyrighted content, including historical newspaper articles. Is there anything else I can help you with?

4. Evaluation

a) WER and CER
b) Qualitative review

The Dataset

The Scotsman is a daily newspaper that has been published in Edinburgh since 1817. In 2022, the University of Edinburgh purchased a licence for the digitised archive (1817-1950) of the newspaper from the ProQuest Historical Newspapers database. The dataset contains PDF images of each issue and corresponding metadata—including full text extracted through OCR—in XML format.

However, usability of the dataset is limited by high rates of OCR errors, particularly in the older issues. Many of the scanned PDF images are poor quality or low resolution; degraded, faded, or missing text is common.

Foreign Intelligence.

FRANCE.

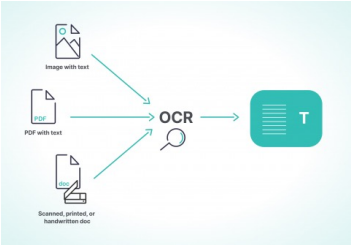
(From the Moniteur of Thursday, Dec. 28.)

OFFICIAL PART.

Ordonnance of the King Louis, by the grace of God, King of France and Navarre. To all to whom these presents shall come, greeting. Considering art. 50 of the Constitutional chartre—considering the laws of 5th February, 10th March, 1816; 20th June, 1820—considering our ordonnances of 4th September and 11th October 1820, on the report of our ministers, Secretary of State

OCR and OCR Errors

Optical Character Recognition (OCR) is a technology that converts scanned images of text into machine-readable digital texts a character at a time. Machine-readable texts are **searchable**, are **compatible with accessibility tools**, and **can be analysed computationally**. OCR is widely used for digitising library and archive collections.



OCR errors occur when layouts or characters are deleted, inserted, or misidentified, such as when the scanner does not recognise a column boundary or interprets the letter 'm' as 'n'.

Insertion
Deletion
Substitution

the North British Railway → tho North Britsh Bailwayy

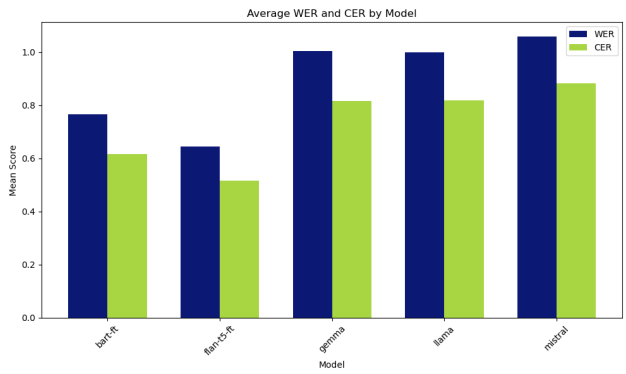
In research, OCR errors can lead to misinterpretation of data, reduced reproducibility, and compromised results. OCR error detection and correction is therefore a critical step in many digital humanities projects.

Measuring OCR Accuracy

OCR errors are commonly assessed using quantitative formulas such as **word error rate** (WER) and **character error rate** (CER). WER measures word-level errors, while CER measures character-level errors. Both formulas measure results based on counting substitutions (S), deletions (D), and insertions in relationship to the total number of words or characters in a text.

$$WER = \frac{S_w + D_w + I_w}{N_w} \qquad CER = \frac{S + D + I}{N}$$

Results



- On average, the fine-tuned LLMs **outperformed** the base LLMs for zero-shot post-OCR error correction
- However, the mean error rates are still well above **the 5-10% threshold** for compromising downstream text analysis
- These results are based on WER and CER scores alone, but qualitative analysis raises additional questions
  - LLMs modernised historical language and grammar; replaced Scots and regional phrases with standard British English
  - Where text is degraded, how do we balance restoration and conservation?
- Future work will aim to further improve performance of post-OCR error correction by trialling full-scale fine-tuning of larger models and multimodal image-to-text models